

Fullstack Developer Course

Interview Preparation Guide

Prepared for: **Nuom Sakada**

Frontend • Backend • Database • Personal Q&A

Part 1 — Personal / General Questions

Answer naturally, in your own words — don't memorize word for word. Speak slowly and smile.

1. Please introduce yourself.

ខ្ញុំបាទឈ្មោះនឿម សក្កា បច្ចុប្បន្នជានិស្សិតនៅវិទ្យាស្ថាន ISTAD ផ្ទាល់។ ខ្ញុំមានស្រុកកំណើតនៅខេត្តសៀមរាប និងជាអតីតសិស្សនៃវិទ្យាល័យអង្គរខេត្តសៀមរាប។

English version (in case they ask in English):

Hello, my name is Nuom Sakada. I am currently a student at ISTAD Institute. I am 19 years old and I come from Siem Reap province. I graduated from Angkor High School in Siem Reap.

2. Why did you choose this institute (ISTAD)?

I chose ISTAD because it focuses specifically on IT and software development, which is exactly the field I want to grow in. ISTAD also offers scholarship courses, which gave me the opportunity to study even though my family could not pay a high tuition fee. Another reason is that ISTAD gives students a chance to continue studying abroad after finishing the program, which matches my long-term goal of growing further in this field.

3. Why are you interested in this course (Fullstack Development)?

I became interested in IT after joining a STEM program in high school, where I was introduced to web coding for the first time. I enjoyed how I could build something visible and useful just by writing code. Fullstack development interests me because I get to work on both what the user sees (frontend) and what happens behind the scenes (backend), so I can build a complete application from start to finish by myself.

4. What are the advantages and disadvantages of this institute?

Advantages:

- Curriculum is focused and practical, built specifically for IT/software careers
- Scholarship opportunities make quality education accessible
- Hands-on projects, so students practice real coding instead of only theory
- Pathway to study further abroad after the program

Disadvantages:

- The pace can be intense since a lot of technology is covered in a limited time
- As a growing institute, some facilities or class sizes may still be improving

Tip: Always end a "disadvantage" answer on a positive note, like above — it shows maturity.

5. What would you tell the new generation of students at this institute?

I would tell them to stay consistent and not be afraid of making mistakes while coding, because that is how everyone learns. I would also encourage them to practice every day, even a little, and to take advantage of the scholarship and mentorship opportunities ISTAD provides, since these chances can really change their future.

6. How did you learn about this institute? (friend / social media / family?)

I first learned about ISTAD through [choose the true one: a friend who was already studying there / social media posts about the scholarship program / a family member who recommended it]. After researching more on their website, I felt confident it was the right place to start my IT journey.

Fill in with whichever is actually true for you before the interview.

7. What is Fullstack (development)?

Fullstack development means being able to work on both the frontend and the backend of an application. The frontend is the part users see and interact with in their browser, while the backend handles the logic, database, and server side of the application. A fullstack developer can build a complete web application from the user interface all the way down to the database.

8. What are your strengths and weaknesses?

Strength: I am a fast learner and enjoy solving problems step by step until I fully understand them. **Weakness:** Sometimes I spend a bit too long perfecting small details, so I am learning to manage my time better and know when "good enough" is fine for now.

9. Where do you see yourself in the next few years?

I see myself becoming a confident fullstack developer, able to build real projects independently, and continuing to learn new technologies. I would also like to have the opportunity to study further, whether through more advanced courses or possibly abroad, to grow my skills even more.

10. Why should we accept you into this course?

I am serious about learning, I already have a foundation from my STEM background in high school, and I am ready to commit time and effort to practice every day. I believe with the right guidance from ISTAD, I can grow quickly and become a skilled developer.

Part 2 — Frontend Questions

HTML, CSS, JavaScript, DOM, Async/Sync, Fetch API, Next.js

1. What is Frontend?

Frontend is the part of a website or application that users see and interact with directly in their browser — the layout, text, images, buttons, and design. It is built mainly with HTML, CSS, and JavaScript.

2. What is HTML and what is it used for?

HTML (HyperText Markup Language) is the standard markup language used to create the structure and content of a web page, such as headings, paragraphs, images, and links.

3. What is CSS and what is it used for?

CSS (Cascading Style Sheets) is used to style HTML elements — controlling colors, fonts, spacing, layout, and how the page looks on different screen sizes.

4. What is JavaScript?

JavaScript is a programming language that makes web pages interactive. It can respond to clicks, update content without reloading the page, validate forms, and communicate with a server.

5. What is the DOM (Document Object Model)?

The DOM is a tree-like structure that represents the HTML elements of a page in memory. JavaScript uses the DOM to find, change, add, or remove elements on the page dynamically.

6. What is the difference between synchronous and asynchronous programming?

Synchronous code runs line by line, and each line must finish before the next one starts. **Asynchronous** code allows a task (like fetching data) to run in the background so the rest of the program can continue without waiting, and the result is handled later (using callbacks, promises, or `async/await`).

7. What is the Fetch API?

The Fetch API is a built-in JavaScript feature used to make HTTP requests to a server, for example to get or send data (often in JSON format) without reloading the page. It returns a Promise.

8. What is a URL and what are its parts?

A URL (Uniform Resource Locator) is the address used to find a resource on the internet. It usually contains: protocol (`https`), domain (`example.com`), path (`/products`), and sometimes query parameters (`?id=5`).

9. What is Next.js?

Next.js is a React framework used to build fast, production-ready web applications. It adds useful features on top of React such as server-side rendering, static site generation, file-based routing, and built-in API routes.

10. What is the difference between let, const, and var in JavaScript?

var is function-scoped and old-style; let and const are block-scoped (modern). let allows the value to change later, while const means the variable cannot be reassigned.

11. What is a Promise in JavaScript?

A Promise is an object representing the eventual result of an asynchronous operation. It can be in one of three states: pending, fulfilled (success), or rejected (failure).

12. What is responsive design?

Responsive design means a website automatically adjusts its layout to look good on different screen sizes, such as mobile phones, tablets, and desktops — usually done with CSS media queries or flexible layouts like Flexbox/ Grid.

13. What is the difference between GET and POST in a Fetch request?

GET is used to request/read data from a server, and data is usually sent through the URL. **POST** is used to send new data to the server (like submitting a form), and the data is sent in the request body.

Part 3 — Backend Questions

OOP, Java, Collections, Spring, Spring Boot, HTTP, CRUD, REST API, JSON

1. What is Backend?

Backend is the server-side part of an application that users don't see directly. It handles business logic, communicates with the database, processes requests, and sends responses back to the frontend.

2. What is OOP (Object-Oriented Programming)?

OOP is a programming style based on "objects," which contain data (fields) and behavior (methods). Its four main principles are: **Encapsulation**, **Inheritance**, **Polymorphism**, and **Abstraction**.

3. What is Java and why is it popular for backend development?

Java is an object-oriented programming language known for being reliable, portable ("write once, run anywhere"), and having a large ecosystem of frameworks like Spring, making it a popular choice for building backend systems.

4. What is a Collection in Java?

A Collection is a framework in Java used to store, organize, and manipulate groups of objects. Common types include **List** (ordered, allows duplicates), **Set** (no duplicates), and **Map** (key-value pairs).

5. What is the difference between List, Set, and Map?

List keeps elements in order and allows duplicate values (e.g., ArrayList). **Set** does not allow duplicate values (e.g., HashSet). **Map** stores data as key-value pairs, where each key is unique (e.g., HashMap).

6. What is Spring Framework?

Spring is a popular Java framework used to build enterprise applications. It provides tools for dependency injection, managing objects, and integrating with databases, making backend development faster and more organized.

7. What is Spring Boot, and how is it different from Spring?

Spring Boot is built on top of Spring, but it simplifies setup by providing auto-configuration, an embedded server, and ready-made "starter" dependencies, so developers can build and run applications quickly without a lot of manual configuration.

8. What is HTTP?

HTTP (HyperText Transfer Protocol) is the protocol used for communication between a client (browser) and a server. It defines how requests and responses are formatted and transmitted over the web.

9. What are the common HTTP methods, and what does CRUD mean?

CRUD stands for Create, Read, Update, Delete — the four basic operations of persistent data. They map to HTTP methods: **POST** (Create), **GET** (Read), **PUT/PATCH** (Update), and **DELETE** (Delete).

10. What is a REST API?

REST (Representational State Transfer) is an architectural style for designing web APIs. A REST API uses standard HTTP methods and URLs to let clients access and manipulate resources, usually exchanging data in JSON format.

11. What is JSON?

JSON (JavaScript Object Notation) is a lightweight, text-based data format used to exchange data between a client and a server. It is easy for both humans and machines to read, using key-value pairs similar to a Java Map or JavaScript object.

12. What is an API?

An API (Application Programming Interface) is a set of rules that allows one piece of software to communicate with another — for example, allowing a frontend application to request data from a backend server.

13. What is Dependency Injection in Spring?

Dependency Injection is a design pattern where Spring automatically creates and provides ("injects") the objects a class needs, instead of the class creating them itself. This makes code more flexible and easier to test.

14. What is the difference between an interface and a class in Java?

A **class** is a blueprint for creating objects with actual implemented behavior. An **interface** only defines method signatures (a contract) without implementation, and a class can "implement" an interface to promise it will provide that behavior.

Part 4 — Database Questions

SQL, NoSQL, Keys, Joins, Relational Model, Normalization

1. What is a Database?

A database is an organized collection of data that can be easily accessed, managed, and updated. It stores information in a structured way, usually using tables (in relational databases).

2. What is SQL?

SQL (Structured Query Language) is a language used to communicate with relational databases — to create tables, and to insert, read, update, and delete data.

3. What is the difference between SQL and NoSQL databases?

SQL databases (like MySQL, PostgreSQL) store data in structured tables with fixed schemas and relationships. **NoSQL** databases (like MongoDB) store data in flexible formats such as documents, key-value pairs, or graphs, and don't require a fixed schema.

4. What is a Primary Key?

A Primary Key is a column (or set of columns) in a table that uniquely identifies each row. It cannot contain duplicate values or be null.

5. What is a Foreign Key?

A Foreign Key is a column in one table that refers to the Primary Key of another table. It is used to create a link/relationship between two tables.

6. What is a relational database?

A relational database organizes data into tables made of rows and columns, where tables can be linked to each other through keys (primary and foreign keys), allowing data to be connected and queried together.

7. What is an INNER JOIN?

An INNER JOIN returns only the rows that have matching values in both tables being joined. If there's no match, the row is excluded from the result.

8. What is a LEFT JOIN?

A LEFT JOIN returns all rows from the left (first) table, and the matching rows from the right table. If there is no match, the result shows NULL for the right table's columns.

9. What is a RIGHT JOIN?

A RIGHT JOIN returns all rows from the right (second) table, and the matching rows from the left table. If there is no match, the left table's columns show NULL.

10. What is a FULL (OUTER) JOIN and a CROSS JOIN?

A **FULL JOIN** returns all rows from both tables, matching where possible and filling with NULL where there's no match on either side. A **CROSS JOIN** returns every possible combination of rows from both tables (a Cartesian product), without needing a matching condition.

11. What is normalization?

Normalization is the process of organizing data in a database to reduce redundancy (duplicate data) and improve data integrity, usually by splitting large tables into smaller, related tables.

12. What is 1NF (First Normal Form)?

1NF requires that each column in a table holds only atomic (single, indivisible) values, and each record is unique — no repeating groups or multiple values in one cell.

13. What is 2NF (Second Normal Form)?

2NF requires that a table is already in 1NF, and that every non-key column depends on the whole primary key, not just part of it (this mainly matters when a table has a composite primary key).

14. What is 3NF (Third Normal Form)?

3NF requires that a table is already in 2NF, and that every non-key column depends only on the primary key, not on another non-key column (no "transitive dependency").

15. What is an index in a database, and why is it useful?

An index is a special structure that speeds up how quickly the database can find rows in a table, similar to an index in a book. It makes searching faster, but adds a small cost when inserting or updating data.

16. What is a one-to-many relationship?

A one-to-many relationship means one record in a table can be related to many records in another table, but each of those records relates back to only one record in the first table — for example, one User can have many Orders.

Quick tips before the interview

- Speak slowly and confidently — it's okay to pause and think before answering.
- If you don't fully remember an answer, explain the concept in simple words rather than staying silent.
- For the personal questions, practice out loud a few times so it sounds natural, not memorized.
- Bring a positive attitude — interviewers often care as much about your attitude as your technical answers.